Problem Set #5: Classical Mechanics (PH31207)

August 31, 2026

Variational Principle

1. Consider the functional $I[y] = \int_a^b F\left(y, \dot{y}, \ddot{y}, x\right) dx$.

   (a) Show that the corresponding Euler equation is
   $$\frac{\partial F}{\partial y} - \frac{d}{dx}\frac{\partial F}{\partial \dot{y}} + \frac{d^2}{dx^2}\frac{\partial F}{\partial \ddot{y}} = 0,$$
   where y and $\dot{y}$ are prescribed at the two end-points.

   (b) What order differential equations is this? How many constants does one need to fully specify a solution of such equation? How many independent boundary conditions are needed? Argue that you have enough such conditions.

2. The objective of this problem is to show that $T+V$ is conserved as a consequence of Lagrange's equation as long as L *does not contain explicit time dependence.* For this, you could use the Euler's homogeneous function theorem, which says:
   $$kf(x_1, \ldots, x_n) = \sum_{i=1}^{n} x_i \frac{\partial f}{\partial x_i}(x_1, \ldots, x_n),$$
   where f is a homogeneous function of n real variables of degree k.

   (a) Test the validity of the above theorem for $f_1(x, y) = x^2 + xy + y^2$ and $f_2(x, y) = x^4 + y^4$.

   (b) Apply this theorem to kinetic energy. Then calculate $\frac{dT}{dt}$.

   (c) Now calculate $\frac{dL}{dt}$.

   (d) Using the expressions of $\frac{dL}{dt}$ and $\frac{dT}{dt}$, show that $\frac{dT}{dt} + \frac{dV}{dt} = 0$ as long as V does not depend on $\dot{q}_k$'s.

   (e) For scleronomic holonomic constraints, we have $T = \frac{1}{2}\sum_{j,l} M_{jl}(\vec{q})\dot{q}_j\dot{q}_l$, where M_{jl} is symmetric. Using above show that $p_i = \sum_k M_{ik}\dot{q}_k$, where p_i is the ith generalized momentum.

   (f) Using above, show that $\sum_i p_i\dot{q}_i = 2T$.

   (g) If L does not explicitly depend on t, what can you comment about $\sum_i p_i\dot{q}_i - L$?